Seminar 1

Putting a price on time and risk

Complements Lecture 1: the three pillars, in practice

The big picture: why this matters

Almost every decision in finance, saving, borrowing, buying a house, valuing a company, comes down to one question: what is a future pound worth today, and how much extra do I need for taking a risk?

1. The power of starting early

Save £200 a month from age 22 and, at about 6% a year, you reach roughly £480,000 by 65.

Start at 32 instead and you get about £250,000.

You paid in only £24,000 more by starting ten years earlier, yet you retire with about £230,000 more. That is compounding.

2. When rates jump, values fall

From 2021 to 2023 the Bank of England raised Bank Rate from 0.1% to 5.25%.

Suddenly the same future cash flows were discounted much harder. UK long-dated government bonds had one of their worst years on record and house prices stalled.

Higher discount rate, lower value today. The maths in this seminar moved markets.

Time and risk have a price. Finance is the business of computing it, and that starts today.

Where this fits, and today's aim

From Lecture 1 (the ideas)

- Price is how society allocates resources; the base rate is the anchor, every other rate is base + a risk premium.
- The three pillars: time value of money, risk management, asset pricing.
- Riskier assets must offer a higher expected return.

This seminar (what you'll DO)

- move money across time by compounding and discounting;
- turn a set of cash flows into an NPV and make an accept / reject decision;
- explain why a riskier asset needs a higher expected return, with numbers.

The anchor: the price of credit

Every return you will ever compute is quoted against one number: the risk-free base rate.

Base rate (risk-free)	the Bank of England anchor
+ term premium	longer money costs more
+ credit premium	riskier borrower, higher rate
= the rate on a loan or bond	what the borrower actually pays

So: when the base rate rises, the discount rate on every asset rises, and asset prices fall. That is the whole intuition behind bonds falling when rates rise.

Compounding: money grows forward

FV = PV x (1 + r)^n years — **future value = today's amount grown at rate r for n years**

Worked example (do it on the board):

Put £100 in an account paying 5% a year. What is it worth in 3 years?

Year 1: 100 x 1.05 = £105.00

Year 2: 105 x 1.05 = £110.25

Year 3: 110.25 x 1.05 = £115.76

Or in one step: 100 x 1.05^3 = £115.76

The £15.76 is 'interest on interest', that is what compounding means.

Discounting: money seen from today

PV = FV / (1 + r)^n **discounting is compounding run in reverse**

Worked: what is £1,000 received in 3 years worth today, at 8%?

PV = 1,000 / 1.08^3

= 1,000 / 1.2597

= £793.83

Your turn (2 min)

What is £500 received in 2 years worth today at 10%?

Write PV = 500 / 1.10^2

NPV: turn cash flows into a decision

A project costs £1,000 now and pays £600 at the end of year 1 and £600 at the end of year 2. Discount rate 10%. Do it?

Time	Cash flow	Discounted at 10%
Now	-£1,000	-£1,000.00
Year 1	+£600	600 / 1.10 = £545.45
Year 2	+£600	600 / 1.21 = £495.87
NPV		**+£41.32**

Decision rule

NPV > 0 accept

NPV < 0 reject

Here NPV = +£41.32, so accept: it earns more than the 10% we could get elsewhere.

The £41.32 is value created today, over and above the 10% opportunity cost of the money.

Asset pricing in action: price a bond

A bond pays a £5 coupon at the end of each year for 3 years, and returns its £100 face value at the end of year 3. What is it worth today?

At a 5% discount rate

Year 1: 5 / 1.05 = £4.76

Year 2: 5 / 1.05^2 = £4.54

Year 3: 105 / 1.05^3 = £90.70

Price = £100.00 (at par)

Now rates rise to 7%

5 / 1.07 + 5 / 1.07^2 + 105 / 1.07^3

Price = £94.76

Same bond, higher discount rate, lower price.

A bond is just discounted cash flows. When the coupon equals the rate it prices at par; when rates rise its price falls, the inverse rate-price link from the start of the seminar.

Risk and return, with numbers

Expected return = the probability-weighted average of the possible outcomes

A share has a 50% chance of returning +20% and a 50% chance of returning -10%.

Expected return = 0.5 x 20% + 0.5 x (-10%) = 10% - 5% = +5%.

The actual return will be +20% or -10%, never exactly +5%. Expected is not guaranteed.

Why risk needs reward:

- nobody holds a risky asset unless its expected return beats the risk-free rate;
- that gap is the risk premium, and it grows with how risky the asset is.

In your groups: where would the money go?

Your group has £10,000 to invest for 10 years. Using today's ideas, agree how to split it across a savings account, bonds, a global shares fund, and crypto.

Think about

Time: what does inflation do to cash over 10 years? Risk and return: which options should pay more, and why? Diversification: should it all go in one place?

Decide and defend

Agree one split as a group. Be ready to say roughly what return you are hoping for, and the single biggest risk you are taking.

There is no one right answer. A strong answer weighs the risk-return trade-off, and uses time and diversification on purpose.

Three traps to avoid in the exam

1

Wrong number of periods

Discount £500 in 2 years as 500/1.10^2, not 500/1.10. Count the years to the cash flow.

2

Adding cash flows without discounting

£600 next year plus £600 the year after is NOT £1,200 today. Discount each one first.

3

'Higher risk means higher return'

Higher risk means a higher EXPECTED return, the required reward. The realised return can still be a loss.

Exam-style question

At a discount rate of 10%, which is worth the MOST today?

A) £100 received today

B) £110 received in one year

C) £121 received in two years

D) they are all worth the same today

Answer: D. 110/1.10 = 100, and 121/1.10^2 = 100, so all three have the same present value.

Take these away

- Compounding grows money forward, FV = PV(1+r)^n; discounting brings it back, PV = FV/(1+r)^n.
- NPV discounts every cash flow to today and nets them off; NPV > 0 means accept.
- The discount rate is the opportunity cost of money: the risk-free rate plus a risk premium.
- Expected return is a probability-weighted average, not a promise.

Before Seminar 2: make sure you can discount a single cash flow and compute a two-year NPV without notes.